

DIFFERENTIAL EQUATIONS

Lecture 6 — Non-Exact ODEs & Integrating Factors

When an ODE is Not Exact

Recall from Lecture 5 that $M dx + N dy = 0$ is exact when $\partial M/\partial y = \partial N/\partial x$. When this test **fails**, the equation cannot be solved directly using the exact method.

Definition: Integrating Factor (IF)

An **integrating factor** is a function μ which, when the entire equation is multiplied by it, makes the resulting equation **exact**. It is sometimes called the *magic factor* because after multiplying, the exact solution method applies directly.

$$M dx + N dy = 0 \quad \xrightarrow{\times \mu} \quad \mu M dx + \mu N dy = 0 \quad (\text{now exact})$$

The challenge is finding μ . In this course, μ is assumed to depend on either x alone or y alone, which leads to two rules.

The Two Rules for Finding the IF

Rule 1: IF is a function of x only

Compute:

$$\frac{M_y - N_x}{N}$$

If this simplifies to a **function of x alone** (no y remaining), then:

$$\text{IF} = e^{\int \frac{M_y - N_x}{N} dx}$$

Rule 2: IF is a function of y only

Compute:

$$\frac{N_x - M_y}{M}$$

If this simplifies to a **function of y alone** (no x remaining), then:

$$\text{IF} = e^{\int \frac{N_x - M_y}{M} dy}$$

Which rule to try first? Always try Rule 1 first. If $(M_y - N_x)/N$ still contains y , it won't simplify to a function of x alone — try Rule 2. One of the two will always work for the types of problems in this course.

Note the sign difference: Rule 1 uses $M_y - N_x$ in the numerator, while Rule 2 uses $N_x - M_y$ (the opposite sign). Do not mix these up.

Full Solution Procedure

Complete Method: Non-Exact ODE

1. Write the equation in the form $M dx + N dy = 0$.
2. Test: compute $\partial M/\partial y$ and $\partial N/\partial x$. If they are equal, it is already exact — use the Lecture 5 method directly.
3. If not exact, try Rule 1: check if $(M_y - N_x)/N$ is a function of x only.
4. If Rule 1 fails, try Rule 2: check if $(N_x - M_y)/M$ is a function of y only.
5. Compute the IF using the appropriate rule.
6. Multiply the entire equation by the IF to get a new M^* and N^* .
7. Verify that the new equation is exact: $\partial M^*/\partial y = \partial N^*/\partial x$.
8. Apply the Lecture 5 solution formula to the new exact equation:

$$\int M^* dx + \int [\text{terms of } N^* \text{ free from } x] dy = C$$

Worked Examples**Example 1: $-y dx + x dy = 0$** **Identify M and N :**

$$M = -y, \quad N = x$$

Test for exactness:

$$\frac{\partial M}{\partial y} = -1 \quad \frac{\partial N}{\partial x} = 1 \quad -1 \neq 1 \quad \Rightarrow \quad \text{NOT exact}$$

Try Rule 1:

$$\frac{M_y - N_x}{N} = \frac{-1 - 1}{x} = \frac{-2}{x}$$

This is a function of x only. ✓**Find the IF:**

$$\text{IF} = e^{\int -2/x dx} = e^{-2 \ln x} = e^{\ln x^{-2}} = x^{-2} = \frac{1}{x^2}$$

Multiply through by $1/x^2$:

$$-\frac{y}{x^2} dx + \frac{1}{x} dy = 0 \quad \Rightarrow \quad M^* = -\frac{y}{x^2}, \quad N^* = \frac{1}{x}$$

Verify exactness:

$$\frac{\partial M^*}{\partial y} = -\frac{1}{x^2} \quad \frac{\partial N^*}{\partial x} = -\frac{1}{x^2} \quad \checkmark \text{ EXACT}$$

Integrate M^* w.r.t. x :

$$\int \left(-\frac{y}{x^2}\right) dx = \frac{y}{x}$$

Terms of N^* free from x : $N^* = 1/x$ contains x . Nothing to add.**General solution:**

$$\boxed{\frac{y}{x} = C \quad \text{i.e.,} \quad y = Cx}$$

Example 2: $2x \tan y \, dx + \sec^2 y \, dy = 0$ **Identify M and N :**

$$M = 2x \tan y, \quad N = \sec^2 y$$

Test for exactness:

$$\frac{\partial M}{\partial y} = 2x \sec^2 y \quad \frac{\partial N}{\partial x} = 0 \quad \Rightarrow \quad \text{NOT exact}$$

Try Rule 1:

$$\frac{M_y - N_x}{N} = \frac{2x \sec^2 y - 0}{\sec^2 y} = 2x$$

This is a function of x only. ✓**Find the IF:**

$$\text{IF} = e^{\int 2x \, dx} = e^{x^2}$$

Multiply through by e^{x^2} :

$$2xe^{x^2} \tan y \, dx + e^{x^2} \sec^2 y \, dy = 0 \quad \Rightarrow \quad M^* = 2xe^{x^2} \tan y, \quad N^* = e^{x^2} \sec^2 y$$

Verify exactness:

$$\frac{\partial M^*}{\partial y} = 2xe^{x^2} \sec^2 y \quad \frac{\partial N^*}{\partial x} = 2xe^{x^2} \sec^2 y \quad \checkmark \text{ EXACT}$$

Integrate M^* w.r.t. x :

$$\int 2xe^{x^2} \tan y \, dx = e^{x^2} \tan y$$

(since $\int 2xe^{x^2} \, dx = e^{x^2}$ by recognition, and $\tan y$ is constant w.r.t. x)**Terms of N^* free from x :** $N^* = e^{x^2} \sec^2 y$ contains x . Nothing to add.**General solution:**

$$e^{x^2} \tan y = C$$

Example 3: $(xy - 1) \, dx + (x^2 - xy) \, dy = 0$ **Identify M and N :**

$$M = xy - 1, \quad N = x^2 - xy$$

Test for exactness:

$$\frac{\partial M}{\partial y} = x \quad \frac{\partial N}{\partial x} = 2x - y \quad \Rightarrow \quad \text{NOT exact}$$

Try Rule 1:

$$\frac{M_y - N_x}{N} = \frac{x - (2x - y)}{x^2 - xy} = \frac{-x + y}{x(x - y)} = \frac{-(x - y)}{x(x - y)} = \frac{-1}{x}$$

This is a function of x only. ✓**Find the IF:**

$$\text{IF} = e^{\int -1/x \, dx} = e^{-\ln x} = \frac{1}{x}$$

Multiply through by $1/x$:

$$\left(y - \frac{1}{x}\right) dx + (x - y) dy = 0 \quad \Rightarrow \quad M^* = y - \frac{1}{x}, \quad N^* = x - y$$

Verify exactness:

$$\frac{\partial M^*}{\partial y} = 1 \quad \frac{\partial N^*}{\partial x} = 1 \quad \checkmark \text{ EXACT}$$

Integrate M^* w.r.t. x :

$$\int \left(y - \frac{1}{x} \right) dx = xy - \ln|x|$$

Terms of N^* free from x : In $N^* = x - y$, the term x contains x . The term $-y$ is free. Integrate it:

$$\int (-y) dy = -\frac{y^2}{2}$$

General solution:

$$xy - \ln|x| - \frac{y^2}{2} = C$$

Example 4 (Rule 2): $(2xy^2 + y) dx + (2y^3 - x) dy = 0$

Identify M and N :

$$M = 2xy^2 + y, \quad N = 2y^3 - x$$

Test for exactness:

$$\frac{\partial M}{\partial y} = 4xy + 1 \quad \frac{\partial N}{\partial x} = -1 \quad \Rightarrow \quad \text{NOT exact}$$

Try Rule 1:

$$\frac{M_y - N_x}{N} = \frac{(4xy + 1) - (-1)}{2y^3 - x} = \frac{4xy + 2}{2y^3 - x}$$

Still contains both x and y — Rule 1 fails.

Try Rule 2:

$$\frac{N_x - M_y}{M} = \frac{-1 - (4xy + 1)}{2xy^2 + y} = \frac{-4xy - 2}{y(2xy + 1)} = \frac{-2(2xy + 1)}{y(2xy + 1)} = \frac{-2}{y}$$

This is a function of y only. $\checkmark$

Find the IF:

$$\text{IF} = e^{\int -2/y dy} = e^{-2 \ln y} = y^{-2} = \frac{1}{y^2}$$

Multiply through by $1/y^2$:

$$\left(2x + \frac{1}{y} \right) dx + \left(2y - \frac{x}{y^2} \right) dy = 0 \quad \Rightarrow \quad M^* = 2x + \frac{1}{y}, \quad N^* = 2y - \frac{x}{y^2}$$

Verify exactness:

$$\frac{\partial M^*}{\partial y} = -\frac{1}{y^2} \quad \frac{\partial N^*}{\partial x} = -\frac{1}{y^2} \quad \checkmark \text{ EXACT}$$

Integrate M^* w.r.t. x :

$$\int \left(2x + \frac{1}{y} \right) dx = x^2 + \frac{x}{y}$$

Terms of N^* free from x : In $N^* = 2y - x/y^2$, the term $-x/y^2$ contains x . The term $2y$ is free. Integrate it:

$$\int 2y dy = y^2$$

General solution:

$$x^2 + \frac{x}{y} + y^2 = C$$

Condition for Exactness — General Problem

Example 5: Find b such that $(ax + by) dx + (kx + ly) dy = 0$ is exact

Identify M and N :

$$M = ax + by, \quad N = kx + ly$$

Apply the exactness test:

$$\frac{\partial M}{\partial y} = b \quad \frac{\partial N}{\partial x} = k$$

For the equation to be exact we need:

$$b = k$$

Solution when exact (i.e. when $b = k$):

$$\int (ax + by) dx = \frac{ax^2}{2} + bxy$$

Terms of N free from x : in $N = kx + ly$, the term kx contains x . The term ly is free:

$$\int ly dy = \frac{ly^2}{2}$$

$$\boxed{\frac{ax^2}{2} + bxy + \frac{ly^2}{2} = C}$$

Summary

Concept	Key Point
Not exact	$\partial M/\partial y \neq \partial N/\partial x$ — multiply by an IF to fix it
Rule 1 (IF in x)	If $(M_y - N_x)/N = f(x)$ only, then IF = $e^{\int f(x) dx}$
Rule 2 (IF in y)	If $(N_x - M_y)/M = f(y)$ only, then IF = $e^{\int f(y) dy}$
Sign difference	Rule 1: $M_y - N_x$; Rule 2: $N_x - M_y$ (opposite sign)
After multiplying	Always re-verify exactness before applying the solution formula
Solution	Same exact method from Lecture 5: $\int M^* dx + \int N_{\text{free}}^* dy = C$

End of Lecture 6